



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 14

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Program:

```
#include <stdio.h>
```

```
long long recursivePower(int x, int n)
```

```
{
```

```
    if (n == 0)
```

```
        return 1;
```

```
    if (n % 2 == 0)
```

```
    {
```

```
        long long half = recursivePower(x, n / 2);
```

```
        return half * half;
```

```
    }
```

```
    return x * recursivePower(x, n - 1);
```

```
}
```

```
long long iterativePower(int x, int n)
```

```
{
```

```
    long long result = 1;
```

```
    while (n > 0)
```

```
    {
```

```
        if (n % 2 == 1)
```

```
            result *= x;
```

```

    x *= x;

    n /= 2;
}

return result;
}

int main()
{
    int x = 2, n = 10;

    printf("Name: Dharshana Sivakumar\n");

    printf("Fast Power - Iterative vs Recursive\n\n");

    printf("Iterative Fast Power: %lld\n", iterativePower(x, n));

    printf("Recursive Fast Power: %lld\n", recursivePower(x, n));

    return 0;
}

```

Output:

The screenshot shows a C program being executed in an IDE. The source code is displayed on the left, and the output is shown on the right. The program calculates the fast power of 2 to the power of 10 using both iterative and recursive methods, which both result in 1024.

main.c	Run	Output
<pre> 11 return result; 12 } 13 14 return x * recursivePower(x, n - 1); 15 } 16 17 long long iterativePower(int x, int n) 18 { 19 long long result = 1; 20 21 while (n > 0) 22 { 23 if (n % 2 == 1) 24 result *= x; 25 26 x *= x; 27 n /= 2; 28 } 29 30 return result; 31 } 32 33 int main() 34 { 35 int x = 2, n = 10; 36 37 printf("Name: Dharshana Sivakumar\n"); 38 printf("Fast Power - Iterative vs Recursive\n\n"); 39 40 printf("Iterative Fast Power: %lld\n", iterativePower(x, n)); 41 printf("Recursive Fast Power: %lld\n", recursivePower(x, n)); 42 43 return 0; 44 } </pre>		<pre> Name: Dharshana Sivakumar Fast Power - Iterative vs Recursive Iterative Fast Power: 1024 Recursive Fast Power: 1024 </pre>